



KENYATTA UNIVERSITY
UNIVERSITY EXAMINATIONS 2017/2018
SECOND SEMESTER EXAMINATION FOR THE DEGREE OF BACHELOR OF
SCIENCE IN AEROSPACE ENGINEERING

EAR 102: WORKSHOP PRACTICE & PROCESSES I

DATE: MONDAY 27TH AUGUST 2018

TIME: 2.00 P.M. - 4.00 P.M.

INSTRUCTIONS:

- This paper contains FIVE (5) questions.
- You are required to answer **THREE (3)** questions only.
- Question **ONE** is compulsory.
- Attempt any **ONE** question from section **B** and any **ONE** question from section **C**.
- All questions carry equal marks.
- All dimensions are in millimeters unless otherwise stated.

SECTION A

QUESTION ONE (COMPULSORY) (20 MARKS)

Assume that you are provided with a mild steel flat steel bar of dimensions (6,000 x 50 x 5) mm which is perfectly flat. You are required to produce the female part as shown in Figure 1:

- Describe how you would produce the part in a fitting workshop, clearly mentioning the tools, instruments and equipments utilized at each stage. **All dimensions in mm.** [12 marks]
- State any **FOUR** safety precautions to be taken during the above process. [4 marks]
- Give **TWO** geometrical functions to be tested during the above process, and state tools/equipment used in each case. [4 marks]

SECTION B

QUESTION TWO (20 MARKS)

- (a) Define the following terms as applied in workshop safety:

- | | |
|--------------|----------|
| i. Safety | [1 mark] |
| ii. Accident | [1 mark] |
| iii. Damage | [1 mark] |
| iv. Harm | [1 mark] |

- (b) Use sketches to describe how the following workpiece non-linear (geometrical) functions may be tested in the workshop:-

- Flatness
- Roundness
- Parallelism

[2 marks]
[2 marks]
[2 marks]

- (c) Use sketches to describe the following techniques of material removal by a file stating where each is best suited.

- Draw-filing
 - Cross-filing
- Discuss the operation of sawing.

[3 marks]
[3 marks]
[4 marks]

QUESTION THREE (20 MARKS)

- (a) Explain what is meant by the term "gauge" as applied in measurement. [2 marks]
- (b) List any **THREE** gauges commonly used in production work and state the application of each. [3 marks]
- (c) State **FOUR** grades of slip (block) gauges and outline the application of each. [4 marks]
- (d) With the aid of sketches, describe the procedure of wringing and dismantling gauge blocks. [5 marks]

- (e) Use Figure 2 to determine the gauge blocks to choose in order to obtain the following lengths.

- 67.505 mm
- 42.530 mm
- 32.0825 mm

[2 marks]
[2 marks]
[2 marks]

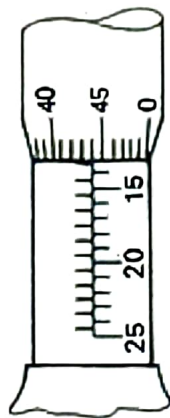
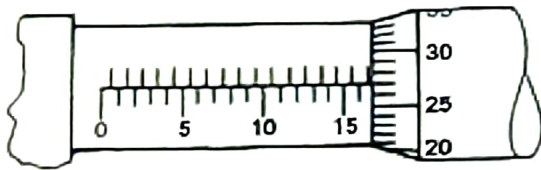
SECTION C

QUESTION FOUR (20 MARKS)

- (a) Using a neat sketch, illustrate the main components of a vernier depth gauge. [5 marks]
- (b) Give the readings of the vernier caliper shown in Figure 3. [2 marks]
- (c) Describe how a reading is measured on a vernier caliper. [3 marks]
- (d) Define the term thread lead stating its importance as applied in micrometers. [3 marks]
- (e) State **TWO** differences between a dot punch and a centre punch. [2 marks]
- (f) List **FIVE** good house keeping factors considered essential in the prevention of accidents. [5 marks]

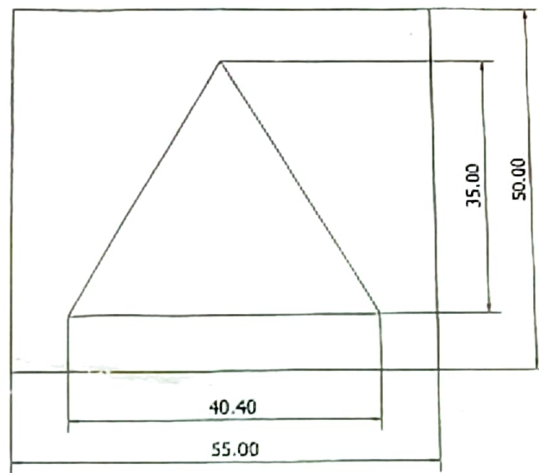
Instructions: Attempt all the Questions

- 1) Define the following terms as applied in workshop safety:
 - (i) Accident, (2 marks)
 - (ii) Hazard and, (2 marks)
 - (iii) Damage. (2 marks)
- 2) List any **TEN** causes of accidents in any workshop (5 marks)
- 3) Give any **TWO** uses of a safety policy in an industry (2 marks)
- 4) Give any **FOUR** aims of a HSW Act 1974 (2 marks)
- 5) Draw a well labelled diagram of a vernier height gauge showing all its components (5 marks)
- 6) Give the readings of the micrometers shown below (3 marks)



(a) (b)

- 7) List **FOUR** types of micrometer screw gauges (2 marks)
- 8) Draw the sketch of a cold chisel and name its various parts. (1 mark)
- 9) Describe the chipping process (3 marks)
- 10) Enumerate **FOUR** points to be kept in mind during the chipping process (6 marks)
- 11) Using neat sketches, describe how to make a rectangular tray (130 x 35 x 20 mm) from a 22 Gauge galvanized iron sheet. (10 marks)
- 12) What do you understand by pinning of files? (2 marks)



FEMALE

THICKNESS=5MM

Figure 1: Question ONE

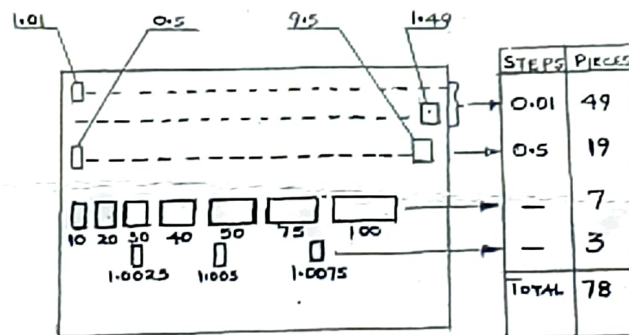


Figure 2: Question THREE

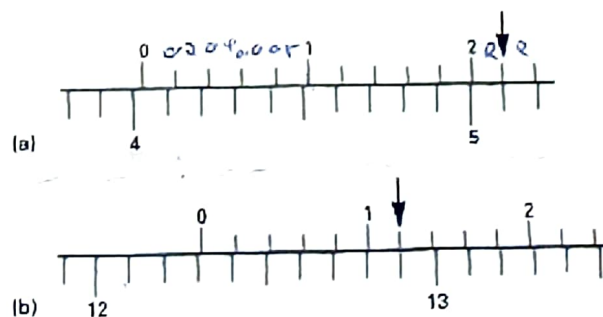


Figure 3: Question FIVE

12.3
24
154

QUESTION FIVE (20 MARKS)

- (a) Give any TWO uses of a safety policy in an industry. [2 marks]
- (b) State FOUR factors considered in selection of metal joining process. [4 marks]
- (c) Describe the soldering process using a neat sketch. [5 marks]
- (d) Define the term brazing as applied in metal joining. [2 marks]
- (e) Differentiate between pan head and flat head rivets using a neat sketch. [4 marks]
- (f) Explain the sequence of using thread taps in internal thread cutting. [3 marks]